

KARGASOK, Russian Federation

WMO: 291220

Lat: **59.0651N**Lon: **80.8325E**Elev: **188**StdP: **14.60**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.1	(c) -36.2	(d) -46.2	(e) 0.4	(f) -41.3	(g) -40.5	(h) 0.5	(i) -35.3	(j) 17.3	(k) 12.0	(l) 16.1	(m) 11.2	(n) 2.5	(o) 270	(p) 0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.9	(c) 82.4	(d) 66.5	(e) 79.5	(f) 65.4	(g) 76.5	(h) 63.9	(i) 69.4	(j) 78.0	(k) 67.4	(l) 75.9	(m) 65.6	(n) 73.8	(o) 5.7	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.7	74.0	64.2	90.7	71.8	62.2	84.2	69.9	33.7	78.1	32.0	76.2	30.6	73.8	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	15.0	13.5	-45.3	87.7	7.2	2.7	-50.5	89.6	-54.7	91.2	-58.7	92.7	-63.9	94.7
			-44.7	72.1	6.4	2.4	-49.3	73.8	-53.1	75.2	-56.7	76.5	-61.3	78.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	-5.8	0.9	18.1	32.2	46.8	61.3	65.1	58.6	47.4	33.2	12.3	0.7
	DBStd	27.19	17.24	15.35	12.70	11.49	10.64	8.29	6.21	6.63	8.18	9.72	16.40	17.20
	HDD50	8166	1729	1374	988	537	191	12	0	7	145	523	1130	1530
	HDD65	12541	2194	1794	1453	983	570	164	78	217	529	984	1580	1995
	CDD50	1260	0	0	0	3	93	349	469	275	67	4	0	0
	CDD65	162	0	0	0	0	6	52	82	20	2	0	0	0
	CDH74	1207	0	0	0	1	74	420	556	142	14	0	0	0
	CDH80	214	0	0	0	0	14	88	94	17	1	0	0	0
Wind	WSAvg	6.5	5.9	6.4	7.2	7.4	7.7	5.9	5.0	5.3	6.0	7.1	7.0	6.6
Precipitation	PrecAvg	19.0	0.7	0.6	0.7	0.9	1.8	2.6	3.0	2.6	2.0	1.6	1.4	1.1
	PrecMax	26.9	2.0	1.4	1.8	2.0	4.8	5.7	5.4	5.4	4.2	3.3	2.3	1.8
	PrecMin	14.0	0.1	0.0	0.1	0.1	0.5	0.4	0.9	0.1	0.2	0.7	0.6	0.3
	PrecStd	3.3	0.4	0.3	0.4	0.4	1.0	1.2	1.3	1.2	0.8	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	33.7	45.2	64.4	81.9	86.4	86.6	82.4	75.7	60.1	41.2	33.0
		MCWB	30.2	31.2	37.1	49.1	61.1	65.9	70.4	67.1	59.7	50.1	38.5	32.0
	2%	DB	26.8	29.8	40.4	57.4	75.2	82.3	82.4	77.9	69.4	54.3	37.0	30.1
		MCWB	25.7	28.0	33.9	46.3	57.8	65.7	67.4	65.3	58.8	47.1	35.1	29.0
	5%	DB	22.0	26.4	37.0	52.4	69.4	79.0	79.8	74.4	65.0	50.4	34.4	27.1
		MCWB	20.9	24.5	32.3	43.5	55.2	64.3	66.7	63.5	56.0	45.4	33.1	25.9
	10%	DB	17.8	21.8	34.4	48.2	63.8	75.2	77.0	70.9	60.4	46.6	32.4	23.2
		MCWB	16.8	20.0	31.1	40.8	52.7	62.8	65.3	62.0	53.7	42.7	31.1	22.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.3	31.8	38.1	51.3	62.6	71.3	72.9	69.6	63.4	51.6	38.7	32.1
		MCDB	31.3	33.4	44.1	61.5	78.0	80.3	80.5	78.3	71.1	56.7	40.4	32.6
	2%	WB	25.6	28.2	34.8	47.3	59.1	67.9	70.5	67.2	59.7	48.4	35.5	29.2
		MCDB	26.7	29.8	38.5	55.8	71.8	77.6	79.2	75.2	67.6	53.2	36.5	30.1
	5%	WB	21.0	24.6	33.1	44.3	56.5	66.0	68.4	64.8	56.7	45.7	33.3	26.0
		MCDB	22.0	26.3	36.4	51.9	67.2	75.9	76.8	72.1	63.4	50.0	34.4	27.0
	10%	WB	16.9	20.0	31.1	41.1	53.8	63.9	66.6	62.7	54.2	42.6	31.3	22.3
		MCDB	17.9	21.7	34.5	47.7	63.1	73.2	74.4	69.4	59.7	46.2	32.4	23.2
Mean Daily Temperature Range	5% DB	MDBR	13.6	15.0	17.3	16.9	16.2	16.2	15.9	15.5	14.1	10.0	11.1	12.9
		MCDBR	12.8	12.3	16.9	21.6	24.9	22.7	20.7	21.6	21.8	15.2	9.0	13.6
		MCWBR	12.4	11.1	12.4	13.0	11.9	9.8	9.3	11.0	12.0	9.6	8.9	13.4
	5% WB	MCDBR	12.6	12.0	14.9	20.6	22.3	19.7	18.3	18.7	19.7	14.2	9.4	13.5
		MCWBR	12.2	11.2	11.8	13.1	11.6	9.6	9.3	10.4	11.6	9.6	9.3	13.4
Clear-Sky Solar Irradiance	taub		0.273	0.308	0.359	0.385	0.393	0.406	0.443	0.411	0.363	0.334	0.289	0.270
	taud		1.982	2.003	1.946	2.072	2.276	2.328	2.196	2.269	2.361	2.260	2.033	1.926
	Ebn at Noon		154	213	236	256	265	263	248	247	242	207	146	108
	Edh at Noon		17	29	41	43	38	36	41	36	28	22	16	13
All-Sky Solar Radiation	RadAvg		176	480	968	1416	1566	1809	1632	1214	767	367	163	98
	RadStd		16	36	61	125	177	180	160	141	101	35	34	10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page